

List of known ADDER v. 1.1.0 bugs and design defects

The Advanced Dimensional Depletion for Engineering of Reactors (ADDER) Software is under continued development at Argonne National Laboratory by the Research and Test Reactor (RTR) Department. In order to satisfy the NQA-1 requirements used for the software, ADDER's development follows a rigorous Software Quality Assurance Plan (SQAP). Rather than being updated every time a modification is made to the software base, the GitHub repository is updated by the developers following a formal release of the software under the SQAP. The timeline of these releases is flexible and is subject to change.

The following list is intended to document known bugs and design defects identified by the ADDER users and developers and currently being addressed as part of the next software release. The users should be aware of these software issues before using the software in order to avoid unintended erroneous results. If you encounter any bug not listed below, please notify the ADDER development team.

Please send any question or comment to adder@anl.gov.

1. Critical Search Operation Convergence Check Bug (Notification 12/11/2025)

Type: Error

Description: During the criticality search operation in ADDER, an early reject check is done for each iteration of a new control group position. If this early reject check passes (i.e., if the 99.5% confidence interval of the calculated k-effective is anywhere within the target range), additional histories are run. Due to a bug introduced in ADDER v1.1.0 development, no additional histories are actually run. Instead, the final convergence check is performed using the same results as used in the early reject check.

This bug causes potentially valid critical positions to be discarded, which can increase the length of the calculation. The critical search operation should still converge if the early reject check produces a k-effective whose 95% confidence interval is entirely within the target range.

How to avoid the issue pending corrective action: Pending a fix, users must be aware that the final results of the critical search operation should not be affected, but the calculation time can increase. However, if the runtime is excessive or if convergence issues occur—potentially due to this bug—users are advised to contact the development team.

2. MCNP Bad Trouble Bug (Notification 5/5/2026)

Type: Design Defect

Description: During an ADDER simulation, MCNP can fail to run successfully in multiple ways. These include non-clean exits (e.g., segmentation faults not logged in the MCNP output) and clean exits because of fatal error or bad trouble as logged in the MCNP output.

ADDER correctly handles non-clean exits and fatal errors; however, does not check for bad trouble. This is generally not a problem, since a run ended by bad trouble typically terminates without the output files or parameters that ADDER needs to continue. However, in certain scenarios, such as the case where an MCNP run terminates due to bad trouble because of an excessive number of particles lost near the end of the MCNP simulation, MCNP can still return outputs with sensical k-effective and nu, but nonsensical tallies. As a result, ADDER can read the MCNP outputs without issuing any error, and the depletion analysis can proceed to the next step based on incorrect results. Also, users may not see messages from screen output, which are not retained by ADDER, and that can indicate a condition that are not explicitly errors or warnings.

How to avoid the issue pending corrective action: Pending a fix, users are recommended to consistently inspect MCNP output files to ensure that the neutronics simulations are executed correctly. If an MCNP simulation terminated because of an error, but ADDER continues running, users should stop the simulation and adjust the MCNP and ADDER inputs accordingly. To obtain all screen output notifications, users can run MCNP standalone. For any questions regarding this issue, users are advised to contact the development team.

3. MCNP Transform Card Jump Bug (Notification 8/21/2026)

Type: Error

Description: In a MCNP input, a rotation matrix can be specified on a transformation card (TR or TRCL) with 3, 5, 6 or 9 entries supplied out of the 9 required. Any omitted entries are entered as jump “j” cards by the user. When ran outside of ADDER, MCNP will automatically calculate entries not supplied (i.e., the jump entries).

ADDER does not correctly handle jump entries included in TR or TRCL cards and can generate an MCNP input with incorrect transformation cards if such jump entries are used. In cases where jump entries are used MCNP calculations ran by ADDER can be incorrect, leading to incorrect results overall.

How to avoid the issue pending corrective action: Pending a fix, users should avoid jump entries in the rotation-matrix portion of `TR`/`TRCL` cards, instead listing out all 9 matrix entries explicitly in the initial MCNP input. For any questions regarding this issue, users are advised to contact the development team.

4. Calculate Volume Operation Bug (Notification 8/24/2026)

Type: Design Defect

Description: ADDER has the ability to calculate material volumes using the calc_volume operation. The current implementation of this operation has two defects.

First, the operation includes the parameter only_depleting, which determines if volumes should be calculated for only depleting materials (True, default value) or all materials (False). Setting this parameter to the value of False has no impact and ADDER will only calculate the volume of depleting materials.

Second, compared to other similar implementations of stochastic volume calculations, for example using MCNP by itself, the ADDER calc_volume operation takes an excessive amount of calculation time for most reactor models.

How to avoid the issue pending corrective action: There is not a way to avoid the issues pending a fix. However, there is no impact on the validity of the volumes calculated for the depleting materials and there is no impact on any other result calculated by the software resulting from these two issues. For any questions regarding this issue, users are advised to contact the development team.